

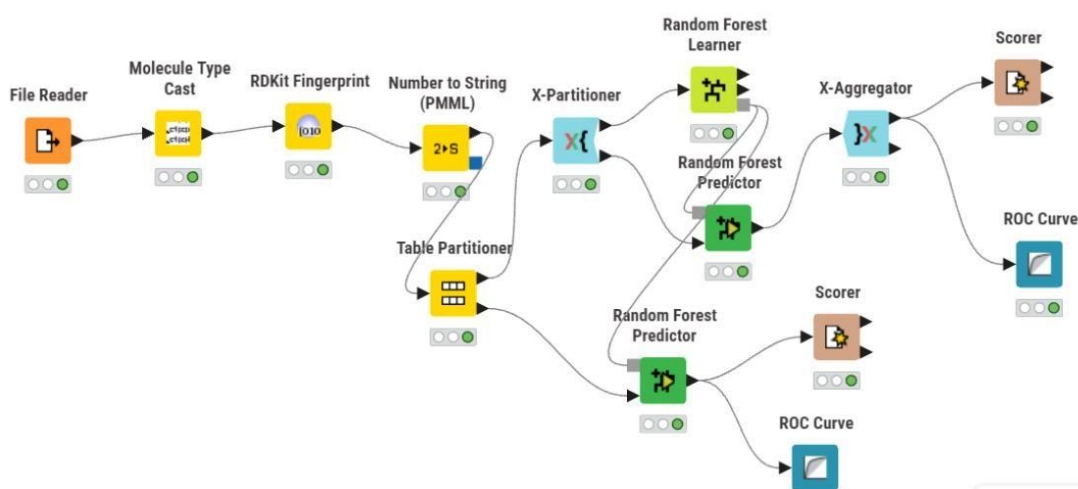
## Computational toxicology and 3R principle-

### Predictive Modelling of Drug-Induced Liver Injury Using Molecular Fingerprints and Random Forest Classification

**Aim:** The aim of this study was to develop and compare predictive machine learning models for drug-induced liver injury (DILI) classification using Random Forest algorithms with two different molecular fingerprint representations named as Morgan fingerprints and MACCS fingerprints for identification of compounds with potential hepatotoxicity at an early stage of drug discovery.

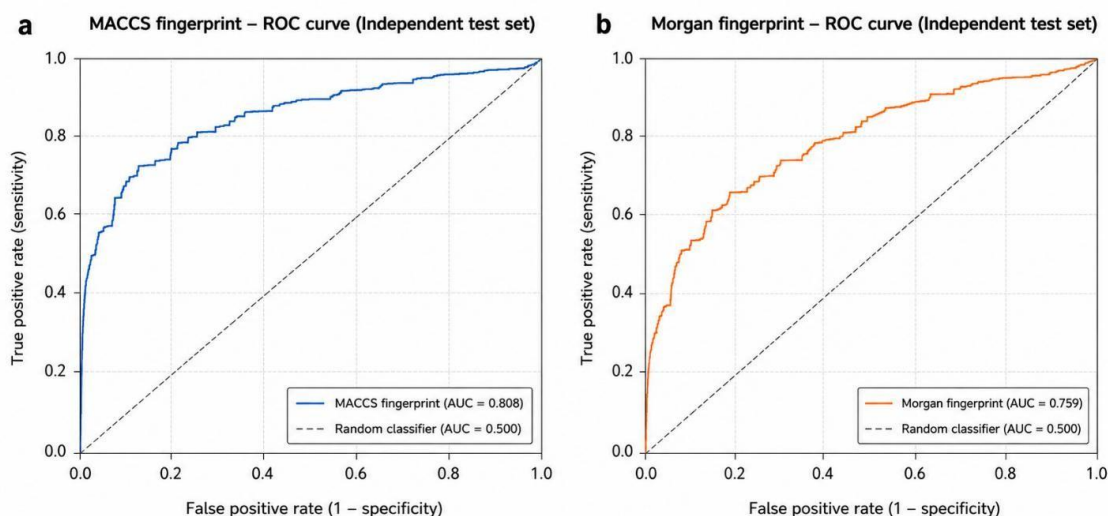
**Task Description:** This study involved developing a predictive machine learning workflow for the classification of drug-induced liver injury (DILI) using molecular structure data. A dataset containing 944 compounds labelled as toxic and non-toxic was used for binary classification analysis. Molecular structures represented in SMILES format were processed using the KNIME analytical platform with RDKit cheminformatics tools to convert chemical strings into molecular objects suitable for computational analysis. Two different molecular fingerprint representations were generated, including Morgan fingerprints (molecular topology and atom connectivity) and MACCS fingerprints (predefined structural subpatterns). These fingerprint descriptors were then used as input features for Random Forest classification models. The task further involved comparing the predictive performance of both fingerprint-based models to determine the most effective molecular representation for early hepatotoxicity prediction in computational drug safety assessment.

**Fig 9. KNIME workflow for predictive machine learning models**



**Results & Discussion:** The Random Forest model developed using Morgan fingerprints demonstrated acceptable predictive performance. During 10-fold cross-validation on the training dataset, the model achieved an accuracy of 77.5%, sensitivity of 44.9%, specificity of 83.5%, and an AUC of 0.759. These findings indicate that the model was more effective at identifying

nontoxic compounds than toxic compounds indicated by relatively lower sensitivity. Evaluation using the independent test dataset showed slightly improved results as accuracy increased to 79.4%, sensitivity reaching 50.0%, specificity rising to 85.4%, and AUC remaining at 0.759. The consistency between training cross-validation and independent test performance suggests stable generalisation without substantial evidence of overfitting.



**Figure 10 |** Receiver operating characteristic (ROC) curves of Random Forest models using different molecular fingerprints. ROC curves were generated on the independent test set. (a) MACCS fingerprint model (AUC = 0.808) shows superior discriminatory ability compared with (b) Morgan fingerprint model (AUC = 0.759).

The dashed line represents the performance of a random classifier (AUC = 0.500).

**Figure 11. Comparison of classification performance of Random Forest models using Morgan and MACCS fingerprints on the independent test set**

| Performance Metric                             | Morgan fingerprint | MACCS fingerprint |
|------------------------------------------------|--------------------|-------------------|
| <b>Confusion Matrix (Independent Test Set)</b> |                    |                   |
| True Positives (TP)                            | 16                 | 15                |
| False Positives (FP)                           | 23                 | 10                |
| True Negatives (TN)                            | 134                | 140               |
| False Negatives (FN)                           | 16                 | 24                |
| <b>Performance Metrics</b>                     |                    |                   |
| Accuracy                                       | 0.759              | <b>0.808</b>      |
| Sensitivity (Recall)                           | <b>0.500</b>       | 0.385             |
| Specificity                                    | 0.854              | <b>0.933</b>      |
| Precision                                      | 0.410              | <b>0.600</b>      |
| F1-score                                       | 0.451              | <b>0.474</b>      |
| ROC-AUC                                        | 0.759              | <b>0.808</b>      |

**Note:** Bold values indicate the best performance for each metric.

The MACCS fingerprint-based Random Forest model showed stronger predictive performance across most evaluation metrics. The MACCS model achieved an accuracy of 81.1%, sensitivity of 58.1%, specificity of 84.3%, and an AUC of 0.808 after training cross-validation. Independent test evaluation further confirmed improved model performance as accuracy reached to 82.0%, sensitivity 60.0%, specificity remaining high at 85.4%, and AUC maintained at 0.808. These findings provides stronger evidence that MACCS fingerprint representation have gained better classification capability and improved discrimination between toxic and non-toxic compounds. Comparison of the two models indicates that the MACCS fingerprint model consistently outperformed the Morgan fingerprint model. The most notable improvement was observed in sensitivity as it increased from 50.0% in the Morgan test model to 60.0% in the MACCS test model. Toxicity prediction prioritises minimising false negatives and this is a crucial improvement that could be addressed (Cavasotto and Scardino, 2022).

Overall, accuracy also improved from 79.4% to 82.0%, while AUC increased from 0.759 to 0.808, demonstrating superior overall discriminatory performance. The results suggest that molecular fingerprint selection has a significant influence on predictive performance in DILI classification. Although both models achieved acceptable classification outcomes, the MACCS-based Random Forest model demonstrated superior effectiveness in identifying toxic compounds while maintaining strong specificity. This suggests that the predefined structural key representation used in MACCS fingerprints could be more suitable for capturing chemical features associated with hepatotoxicity than the broader topological encoding provided by Morgan fingerprint.